ANDRES BARBOZA

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EDUCATION

Ph.D. in Genetics and Genomics 2026 (Expected)
Texas A&M University, College Station, TX
Major GPA: 4.00 ⋄ Committee: Heath Blackmon (Advisor), William Murphy, Alex Keene, Mahul Chakraborty

B.S. in Biochemistry 2022
Arkansas Tech University, Russellville, AR
Major GPA: 3.96 (Summa Cum Laude)

SKILLS

Programming	Python, R, Linux/Unix (Command-line), Bash, C++, HTML/CSS, LaTeX
Code Management	Git/Github, Conda, Docker/Singularity, Snakemake, Nextflow
Genomics	NGS Analysis (bwa, bamtools, GATK, bcftools, vcftools, PLINK, samtools), Genome Assembly (HIFIasm, SPAdes, BUSCO), RNASeq (FastQC, bowtie2, Salmon, DESeq2), GWAS
Data Science	Machine Learning (Computer Vision), Simulations, Data Visualization, Software Development, HPC Cluster Computing
Molecular Biology	gDNA/DNA/RNA extraction, PCR, qPCR, RT-PCR, Gel Visualization/Imaging
Chemistry	UV-Vis Spectrometry, IR Spectroscopy

RESEARCH EXPERIENCE

Graduate Research Assistant 2023 - Present
Texas A&M University

- Focus on genome evolution and population genetics using genomic pipelines and computer simulations.
- Develop scalable computational workflows using Snakemake for population genetics analysis in *Chrysina* beetles.
- Perform *de novo* genome annotation for multiple beetle species (*Woodi*, *Beyeri*, and *Leconti*) utilizing PacBio HiFi sequencing and high-performance computing clusters.
- Conduct mathematical modeling for theoretical evolution and machine learning applications for image recognition.

Pharmacogenetics Clinical Trial Intern Summer 2024
AbbVie

- Compared short-read, long-read, and PharmacoScan sequencing platforms for pharmacogenetics call performance.
- Built and tested pipelines for star-allele calling using WGS and ONT data.

Undergraduate Research Assistant 2021 - 2022
Arkansas Tech University

- Assisted in research activities for laboratory projects on organic fluorescent probes.
- Performed batch titrations on a UV-vis spectrometer and fluorimeter; prepared standard samples.

Summer Research Intern Summer 2021
University of Arkansas for Medical Sciences

- Performed research in osteoblast gene expression involving RT-PCR and cell cultures.
- Wrote a research manuscript for oral presentation.

PUBLICATIONS

ANDRES BARBOZA, GARRETT FISSELL, DANIEL SPALINK, HEATH BLACKMON. "SeqDef: An R package for phylogenetically weighted genomic prioritization across the tree of life." Under Review in *Journal of Evolutionary Biology*.

MEGAN COPELAND, **ANDRES BARBOZA**, PHILIP UGOCHUKWY, HEATH BLACKMON. "Self-compatibility weakens barriers to karyotype change and increases the rate of chromosome evolution in angiosperms." Under Review in *PNAS*.

ANDRES BARBOZA, HEATH BLACKMON. "Drivers of Achiasmatic Meiosis: Sexual Antagonism versus Heteromorphy-Dependent Aneuploidy across Sex-Chromosome Divergence" *G3*. 2025.

MEGAN COPELAND, **ANDRES BARBOZA**, JOSEPH ROMANOWSKI, ZACH ADELMAN, HEATH BLACKMON. "DirectRepeater: R package for annotating direct repeats in genome assemblies." *F1000 Research*. 2025.

ANDRES BARBOZA PEREIRA*, MATTHEW MARANO*, RAMYA BATHALA, RIGOBERTO AYALA ZARAGOZA, ANDRES NEIRA, ALEX SAMANO, ADEKOLA OWOYEMI, CLAUDIO CASOLA. "Orphan Genes are not a Distinct Biological Entity." *BioEssays*. 2024.

RAJIB CHOUDHURY, ARUN K. SHARMA, PRATIKSHYA PAUDEL, PRESTON WILSON, **ANDRES BARBOZA PEREIRA**. "In Situ Generation of a Zwitterionic Charge Transfer Probe for Selective Detection of Human Serum Albumin." *Analytical Biochemistry*. 2022.

PRESENTATIONS

2026

- ANDRES BARBOZA, GARRETT FISSELL, DANIEL SPALINK, HEATH BLACKMON. "SeqDef: An R Package for Phylogenetically Weighted Genomic Prioritization Across the Tree Of Life" Texas A&M University Life Sciences Graduate Recruitment Symposium, College Station, TX. (Poster Presentation)
- ANDRES BARBOZA, GARRETT FISSELL, DANIEL SPALINK, HEATH BLACKMON. "SeqDef: An R Package for Phylogenetically Weighted Genomic Prioritization Across the Tree Of Life" Texas A&M University Biology Student/Post-Doc Research Conference (Poster Presentation)

2025

- ANDRES BARBOZA PEREIRA, ZOYA WANI, HEATH BLACKMON. "Epistatic Capacitance: The Accumulation of Variance in Populations." Southeast Texas Evolutionary Genetics & Genomics, College Station, TX. (Poster Presentation)
- ANDRES BARBOZA PEREIRA, ZOYA WANI, HEATH BLACKMON. "Epistatic Capacitance: The Accumulation of Variance in Populations." Southeast Texas Evolutionary Genetics & Genomics, College Station, TX. (Poster Presentation)
- ANDRES BARBOZA PEREIRA, ZOYA WANI, HEATH BLACKMON. "Epistatic Capacitance: The Adaptive Reservoir of Populations." Texas Genetics Society, College Station, TX. (Poster Presentation)
- ANDRES BARBOZA PEREIRA, ZOYA WANI, HEATH BLACKMON. "Epistatic Capacitance: New Hypothesis on the Source of Adaptation." Texas A&M University Ecology and Evolutionary Biology Seminar, College Station, TX. (Oral Presentation)
- ANDRES BARBOZA PEREIRA, ZOYA WANI, HEATH BLACKMON. "Epistatic Capacitance." Texas A&M University Life Sciences Graduate Recruitment Symposium, College Station, TX. (Poster Presentation)
- ANDRES BARBOZA PEREIRA, ZOYA WANI, HEATH BLACKMON. "Epistatic Capacitance." Texas A&M University Biology Student/Post-Doc Research Conference, College Station, TX. (Poster Presentation)

2024

- ANDRES BARBOZA PEREIRA, HEATH BLACKMON. "Exploring the Evolutionary Dynamics of Achiasmatic Meiosis." Ecological Integration Symposium, Texas A&M University, College Station, TX. (Oral Presentation)
- ANDRES BARBOZA PEREIRA, HEATH BLACKMON. "Exploring the Evolutionary Dynamics of Achiasmatic Meiosis." Texas Genetics Society Meeting, College Station, TX. (Poster Presentation)

- ANDRES BARBOZA PEREIRA, HEATH BLACKMON. "Exploring the Evolutionary Dynamics of Achiasmatic Meiosis." Student Research Week, Texas A&M University, College Station, TX. (Oral Presentation)
- ANDRES BARBOZA PEREIRA, HEATH BLACKMON. "Exploring the Evolutionary Dynamics of Achiasmatic Meiosis." SPRC Conference, Texas A&M University, College Station, TX. (DataBlitz Oral Presentation)
- ANDRES BARBOZA PEREIRA, HEATH BLACKMON. "Exploring the Evolutionary Dynamics of Achiasmatic Meiosis." SPRC Conference, Texas A&M University, College Station, TX. (Poster Presentation)

2023

- ANDRES BARBOZA PEREIRA, HEATH BLACKMON. "Exploring the Evolutionary Dynamics of Achiasmatic Meiosis." G2 Seminar, Texas A&M University, College Station, TX. (Oral Presentation)
- MATTHEW A. MARANO, ADEKOLA OWOYEMI, RIGOBERTO AYALA ZARAGOZA, RAMYA R. BATHALA, ANDRES A. NEIRA, ALEX SAMANO, ANDRES BARBOZA PEREIRA, CLAUDIO CASOLA. "The Evolutionary Origins of Orphan Genes." SMBE Satellite Meeting, Texas A&M University, College Station, TX. (Poster Presentation)
- ANDRES BARBOZA PEREIRA, ALEX C. KEENE, HEATH BLACKMON. "Rest and Activity Behavior of Betta Fish: Insights into Crepuscular Patterns." Gene Editing Symposium, Texas A&M University, College Station, TX. (Poster Presentation)
- ANDRES BARBOZA PEREIRA, ALEX C. KEENE, HEATH BLACKMON. "Rest and Activity Behavior of Betta Fish: Insights into Crepuscular Patterns." Evolution Meeting, Albuquerque, NM. (Poster Presentation)
- ANDRES BARBOZA PEREIRA, JULIETTE STROPE, HEATH BLACKMON. "Fitness relationship between genome fragmentation, environmental stability, and interactions between genes." Biochemistry-Chemistry-Genetics Research Competition, Texas A&M University, College Station, TX. (Poster Presentation)
- ANDRES BARBOZA PEREIRA, JULIETTE STROPE, HEATH BLACKMON. "Fitness relationship between genome fragmentation, environmental stability, and interactions between genes." Texas Genetics Society Meeting, Austin, TX. (Poster Presentation)

2021

- ANDRES BARBOZA PEREIRA, MELDA ONAL. "Osteoblast Differentiation and Function in the Absence of Chaperone-Mediated Autophagy." Arkansas INBRE Summer Program, Little Rock, AR. (Oral Presentation)

UNDERGRADUATE MENTORSHIP

Graduate Student Mentor 2023 - Present
Texas A&M University

- Mentored multiple undergraduate students in computational biology and evolutionary genetics research.
- Guided students through the entire research lifecycle, transitioning them from heavily guided instruction in basic bioinformatics to independent data analysis and problem-solving.
- **Mentee Outcomes:** Successfully prepared undergraduates for advanced scientific careers, with direct mentees moving on to prestigious postgraduate programs, including Kenzie Laird (Ph.D. program at University of British Columbia), Kevin Bolwerk (Ph.D. program at Texas A&M University), Meghann McConnell (Ph.D. program at Clemson University), Zoya Wani (STEGG Post-Baccalaureate Program at University of Houston)

LEADERSHIP & SERVICE

Membership Committee Chair August 2024 - August 2025
GGSA, Texas A&M University

- Reviewing applications, researching new hires, and leading the faculty-student committee.

Seminar Committee Co-Chair August 2023 - August 2024
GGSA, Texas A&M University

- Invited speakers, prepared itineraries, and coordinated meals for student interaction.

TEACHING EXPERIENCE

Graduate Teaching Assistant Spring 2023 - Present
Texas A&M University

- **Introductory Biology II Laboratory (Fall 2025 - Spring 2026):** Lead laboratory sections for undergraduate STEM majors, facilitating foundational biological concepts and experimental techniques.
- **Principles of Genetics Laboratory (Spring 2023 - Spring 2025):** Prepared and delivered weekly lectures, assessed student learning, and guided undergraduates through complex genetic principles.

Undergraduate Teaching Assistant Feb 2019 - May 2022
Arkansas Tech University

- **Organic Chemistry I Laboratory:** Assisted and mentored students in complex laboratory techniques, prepared chemical samples, and graded reports.
- **General Chemistry Laboratory:** Guided STEM majors through foundational chemistry experiments and assessed student coursework.
- **Survey of Chemistry Laboratory:** Facilitated introductory chemistry concepts for diverse student backgrounds and managed laboratory workstations.

Peer Tutor Jan 2020 - Nov 2020
Arkansas Tech University

- Taught Chemistry, Biology, and Calculus, creating tailored lesson plans.

AWARDS

GGSA Teaching Excellence Award 2023

GGSA Conference Award 2023

American Chemical Society Physical Chemistry Achievement Award 2022